Trenitalia
VZI-6

TCMS MMI FRS

Revision	Author	Date	Revision Description
00.01	G. Iusto	30/11/2023	Draft Release for internal Review
00.02	G. Iusto	1/12/2023	According to the design review comments, the following changes were done: 1) The reference to the appendix were replaced by the reference to the VZI-14 fleet. 2) The BT document references were replaced by the HR document references
1.0	**G. Iusto**	**30/09/2024**	**Rilascio in GPLM**

Content

1 Introduction 5

1.1 Train Configuration 5

1.2 Definitions and abbreviations 5

2 System Overview 6

3 System Overview 6

3.1 System Configuration 6

3.2 Normal condition 6

3.3 Failure condition 6

3.4 Failure supervision 7

3.5 TOD safety concept 7

3.5.1 Safe confirm by safety key 8

4 Functional requirements 8

4.1 General 8

4.1.1 Graphical Strategy for TOD 8

4.1.2 Audible message 9

4.1.3 Reset of TOD 9

4.1.4 Back-up TOD 9

4.2 Menu structure and navigation 10

4.2.1 Function Design 10

4.2.2 Requirements 11

4.3 Menus 13

4.3.1 Primary TOD menus 13

4.3.2 Menus for Diagnostics 18

4.3.3 Menus for Instruments 34

4.3.4 Menus for Settings 41

4.3.5 Menus for Test and Maintenance 43

5 Signal Interface 45

5.1 I/O Interface 45

5.1.1 Supplier system 45

5.1.2 TCMS SW 45

5.2 Bus Interface 45

List of figures

Figure 1: Train Architecture 5

Figure 2: TOD-TCMS Architecture 7

List of tables

Table 1: Definition and Mnemonics 5

Table 2: TCMS main components 6

1 Introduction

This document is a function specification for the TCMS Train Operability function for VZI-6 of customer Trenitalia.

1.1 Train Configuration

Each VZI-6 train consists of eight cars in a fixed formation including a driver's cab at each end:

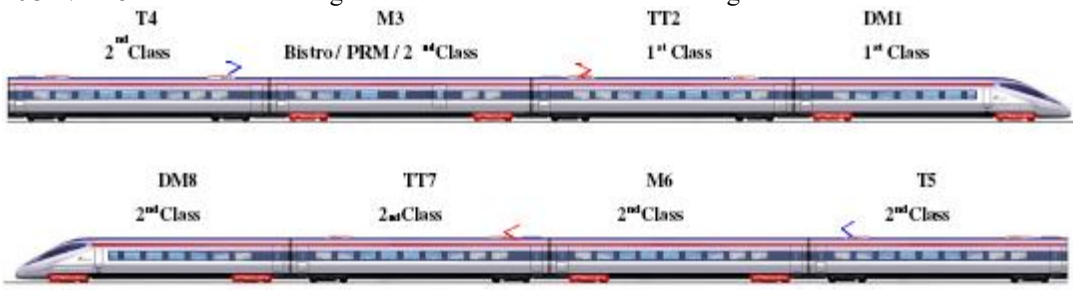


Figure 1: Train Architecture

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
CCU	Central Computing Unit
MDCU	Central Computing Unit – Regular and Safe Diagnostic functions
CCU-R	Central Computing Unit – Regular Operational Functions
CCU-S	Central Computing Unit – Safe Operational Functions
ECN	Ethernet Consist Network
ETB	Ethernet Train Backbone
EVENT	Diagnostic Item that will be stored in the Event database within TCMS system.
GUI	Graphical User Interface
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
ICD	Interface Control Document
MDS	Train Diagnostic System
MDS-DM	Data and Monitoring of the Train Diagnostic System
MDS-DR	Data Recorder of the Train Diagnostic System
IPT	Internet Protocol – Train
MDS	Monitoring and Diagnostic Data System
ODBS	Onboard Diagnostic Database System
PCL	Process Client
PTU	Portable Test Unit
SDTV2	Safe Data Transmission version 2
TCL	Tool Client
TCMS	Train Control and Management System
TOD	Train Operator Display
TSSP	Train Subscription Service Provider
VRS	Version Reporting System
WBS	Work Breakdown Structure

Table 1: Definition and Mnemonics

2 System Overview

The task for the TCMS system is to provide a platform for software-controlled functionalities of the train.

The TCMS was designed to be installed on board railway vehicles and has the following main features:

- control, monitoring and complete management of on-board subsystems;
- management and diagnostics of contactors, relays, valves and all the different components present on the vehicle;
- secure data communication to reduce railway cables;
- recording, archiving and processing of data for diagnostics and maintenance both on board the train and remotely;
- human-machine interface for railway operators, train drivers and maintenance workers.

The TCMS consist of the following components:

Apparecchiatura	Funzione	Note
CCU-R	- Regular Vehicle control Unit	It has to be used for Regular (no-safe) functions)
CCU-S	Safe Vehicle control Unit	It has to be used for Safe functions)
MDCU	Monitoring and Diagnostic Unit	It has to be used for the monitoring and diagnostic functions (safe and no-safe)
DRU	Data Recorder Unit	It has to be used to store signals for further processing durino the off-boards analysis
OBS	On Board Server	It has to be used to share information between the on board and the way side equipment
RIO	Remote IO Unit	It has to be used to manage Input and Output train signals by the communication bus
TOD	Train Operator Display	The TOD is installed in the driver desk and in the crew compartment to implements the MMI Interface; it allows both no-safe and safe functionalities

Table 2: TCMS main components

3 System Overview

The MMI interface consists of the TCMS TODs in the driver's cabs and the crew compartment.

The purpose of the TCMS TOD, hereafter called TOD, is to provide a graphical user interface to the TCMS. This document describes the functional requirements on the TOD. The TOD Picture model describes each menu graphically, see [1].

3.1 System Configuration

There are two TODs located in each cab and one TOD located in the crew compartment in the M3 car. In the cab the primary TOD is in front of the driver and the back-up TOD is located to the left of the driver. The back-up TOD is mounted behind a transparent cover and is not possible to touch. If the primary TOD fails, the driver can replace the primary TOD unit with the back-up TOD unit when the train is at standstill.

3.2 Normal condition

In normal condition mode the primary TOD in the cab is the activated one, i.e. the TOD for the Driver, Maintainer or Crew to interact with. The secondary back-up TOD is in idle mode at this condition.

3.3 Failure condition

In failure condition mode the secondary TOD in the cab, i.e. the backup TOD, is activated. It is then the user's responsibility to move it from the secondary location to the primary location. Once the switch has been performed and the TOD is powered up it is ready to be used.

3.4 Failure supervision

Failure supervision is performed either by the user, seeing that the TOD has become blank or in another abnormal state, or by the lifesign facility between the TOD and train computer.

3.5 TOD safety concept

The TCMS Platform and its safety products, the TOD, are capable of implementing a set of generic safety functions. The VZI6 application will utilize the following generic safety functions:

1) Display safety related data (Secure Data).

2) Data confirmation by a safety key.

In the following the conceptual architecture of the TOD and its connection with the TCMS connection are shown.

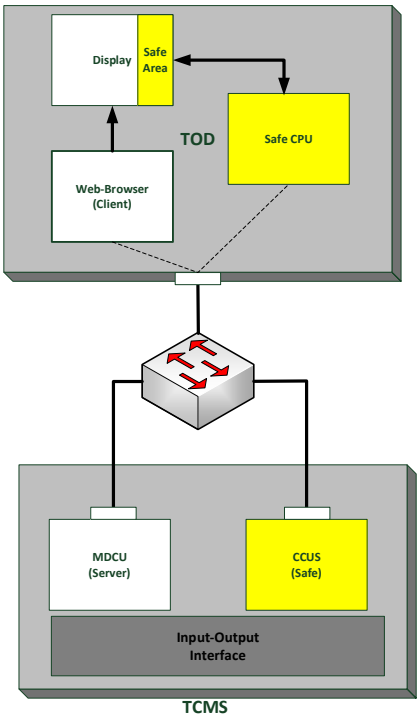


Figure 2: TOD-TCMS Architecture

The TOD receives regular process data from the MDCU and SDTv2 process data (safe data) from the CCUS.

There are two main components in the TOD, the main CPU, and the safety CPU.
The main CPU executes the TOD application:

1) It receives SDTv2 process data and Regular process data over external Ethernet connector.
2) It displays Safe and no Safe components on screen.

The safe CPU perform supervision of the TOD application by the following functions:

1) Extract and validate SDTv2 process data received from the main CPU.
2) Supervise that the main CPU correctly displays Safe Components according to SDTv2 process data.
3) Process button presses on Safe keys, used to confirm SDTv2 process data.

A Safe Area inside the Display is reserved for the Safe Components controlled by the safe CPU.

3.5.1 Safe confirm by safety key

For safe confirm with the safety key, a page configured for safe confirmation shall be displayed. That specific page shall present safe data using display of safe data. On the same page one safety key on the front panel shall be configured as the active confirm button.

4 Functional requirements

4.1 General

4.1.1 Graphical Strategy for TOD

4.1.1.1 *Function Design*

To provide a consistent overall look & feel of the HMI a standard screen layout is used. The standard screen always covers the whole screen, 800 x 600 pixels.
The standard screen layout includes the following fields:

- Header field
- Title field
- Active content field
- Event field
- Footer field

Most menus will incorporate all fields in the standard screen layout. Under specific circumstances, i.e. lack of screen space or desire to show as little information as possible circumstances, certain fields may not be used. This presents an opportunity to enlarge other aspects of the screen, namely the active content field.

The contents of the header field, title field and footer field are specified in the requirements below. The event field is specified in chapter Events. The contents of the active content field is specified for each menu together with any exceptions from the standard screen layout.

4.1.1.2 *Requirements*

ID	2F_10.03.Zefiro-Europe.TRS.11
Description	The screen shall be divided into 3 areas: • Upper part shall present main data (e.g. time, date, train speed) and the train configuration • Middle part shall present the menu data currently selected • Lower part shall present event text and acknowledge feature for Events as well as buttons for navigating on the TOD
Safety level	N/A

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_10.03.Zefiro-Europe.CONCEPT.319	The TOD shall display the time (24 hour format hh:mm:ss) and date (yy-mm-dd) in the header field.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.329	The TOD shall display the title of the menu in the title field.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.330	The TOD shall provide buttons for navigation in the footer field.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_10.03.Zefiro-Europe.TRS.266
Description	The train speed shall be displayed on the TOD.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.327	The TOD shall display the speed in the header field.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 2

4.1.2 Audible message

4.1.2.1 *Function Design*

When an event has occurred that requires the train staff attention there will be an audible message or tone to direct the train staff to the TOD in the crew compartment. An in all car audible message shall be used to get the crew's attention.

4.1.2.2 *Requirements*

ID	2F_10.03.Zefiro-Europe.TRS.181
Description	If an event occurs, requiring train staff attention, this shall be indicated in all cars. An audible message shall be used to direct the train crew to the TOD in the crew compartment.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.392	If an event occurs that requires the train crew's attention an audible message shall be issued in each car.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

4.1.3 Reset of TOD

4.1.3.1 *Function Design*

Restart of the TOD when the TOD has entered non operational safe state is initiated by the CCU-S. When the TOD enters operational safe state the displayed safe areas are blanked.

The TOD will correctly create cyclic, outgoing SDTv2 PD with TOD Safety Processor status at least every 100 ms.
A Safe Train Application which is interested in the status of the TOD safety functions must subscribe to this status data. One part of the TOD Safety Processor status message contains the operational safe state for the TOD Safety Processor. The Safe Train Application on the CCU-S reacts upon the contents of this process data and takes apropriate actions. E.g. initiates restart of the TOD if an Safe Data Area failure has been detected by the TODs' Safety Processor.

The handshaking mechanism between CCU-S and the TOD's Safety Processor is described within SFRS TCMS Operational, see FA020026278 chapter 3.4 Fail Safe State Control.

4.1.4 Back-up TOD

4.1.4.1 *Function Design*

The back-up TOD is located on the backwall in the cab. It is mounted behind a transparent cover so the touch screen can not be operated. It is connected to power and Ethernet network. The back-up TOD is in normal operation in idle mode and not used, it is spare.

The following scenario is foreseen:

When the primary TOD is in failure mode, the back-up TOD is awakened and it will enter the same mode as the primary TOD had before entering failure mode. The back-up TOD also will use the same language as the primary TOD had. Because it is located behind the transparent cover, the functionality will be read only, and it is not possible to interact with the TOD by touching the screen.

The intension is that the Driver will replace the primary TOD with the back-up TOD. The TOD swapping will only be done at standstill. Once the back-up TOD is activated it will have full functionality.

4.1.4.2 *Requirements*

ID	2F_10.03.Zefiro-Europe.TRS.229
Description	In case the primary TOD in the activated cab is failing the back-up TOD shall contain all functionality
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.313	The back-up TOD in the cab shall be in Idle mode if the primary TODs life-sign counter is updating.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.382	If the primary TODs life sign counter is not updating the back-up TOD shall get the same mode as the primary TOD had.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.901	If the primary TODs life sign counter is not updating the back-up TOD shall use the same language as the primary TOD did.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

4.2 Menu structure and navigation

4.2.1 Function Design

The TCMS TOD is accessed when the cab is activated or by touching the TOD dislay. In activated cab access is granted as for user group driver and in crew compartment as for user group crew. In the non activated cab the user always has to login to access the TOD.

If the user wants more access the user needs to login as maintainer using the login menu. If the correct user id is entered access as maintainer is granted.

Notable is that the user can always login in as any of the below user groups by using the login menu on the TOD.
There are three user groups:

- Driver
- Crew
- Maintainer

The Driver has a limited access to the TOD and the Maintainer has a complete access. The principle is that all functionality available for Driver are also available for Maintainer. However the Maintainer can have more functionality on the TOD that is only needed for Maintainers.

The TOD can be used in activated cab, in non activated cab or in the crew compartment. The functionality of the TOD in the non activated and in the crew compartment is read only, i.e. information presentation only, unless there are specific functions related to the location of the TOD that requires control.

Menu structures has been determined by grouping functions appropriately by user group and location of the TOD. The following menu structures exist:

- Driver in activated cab
- Crew in activated cab
- Maintainer in activated cab
- Driver in non activated cab
- Crew in non activated cab
- Maintainer in non activated cab
- Driver in crew compartment
- Crew in crew compartment

- Maintainer in crew compartment

The same menu structure is used as a base for all user groups and locations. The principle is that when a function is not available the corresponding menu is removed keeping the functions geographically consistent in all menu trees.

The menu structure has been designed to be shallow, allowing quick access to functionality that requires it, while not placing too many functions on a single page, avoiding clutter. Shortcuts are available for important and frequently used pages to speed up the navigation process.

Two different types of navigation are present in the TOD application. The two types are manual navigation and forced navigation.

1) Manual navigation is navigation performed by the user, i.e. pressing a navigation button on the TOD display which performs a menu change.
2) Forced navigation is navigation caused by a system state, e.g. when decoupling has been performed the TOD application automatically displays the Decoupling system message menu.

The forced navigation has a priority for deciding of which menu to show.

4.2.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.234
Description	The TOD functionality shall be designed as a structured menu tree where specific system menus and diagnostics menus can be reached depending on user and location of TOD.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.365	The following state modes shall exist for the TOD: · Idle mode · Driver in activated cab · Crew in activated cab · Maintainer in activated cab · Driver in non activated cab · Crew in non activated cab · Maintainer in non activated cab · Driver in crew compartment · Crew in crew compartment · Maintainer in crew compartment	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.366	The TOD shall be in Idle mode when the TOD has got power and no other mode is set.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.367	The TOD shall be in Driver in activated cab mode when the user has activated the cab. The mode shall be left by deactivating the cab or if a user logs in as Maintainer or Crew.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.655	The TOD shall be in Crew in activated cab mode when the user has logged in as Crew in an activated cab. The mode shall be left by deactivating the cab or if a user logs in as Maintainer or Driver	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

2F_10.03.Zefiro-Europe.CONCEPT.368	The TOD shall be in Maintainer in activated cab mode when the user has logged in as Maintainer on the TOD in an activated cab. The mode shall be left by deactivating the cab or if a user logs in as Driver or Crew.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.369	The TOD shall be in Driver in non activated cab mode when the user has logged in as Driver on the TOD in the non activated cab. The mode shall be left after 10 minutes of inactivity, or if the cab is activated, or if a user logs in as Maintainer or Crew.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.656	The TOD shall be in Crew in non activated cab mode when the user has logged in as Crew on the TOD in the non activated cab. The mode shall be left after 10 minutes of inactivity, or if the cab is activated, or if a user logs in as Maintainer or Driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.370	The TOD shall be in Maintainer in non activated cab mode when the user has logged in as Maintainer on the TOD in the non activated. The mode shall be left after 10 minutes of inactivity, if the cab is activated or if a user logs in as Driver or Crew.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.657	The TOD shall be in Driver in crew compartment mode when the user has logged in as Driver on the TOD in the crew compartment. The mode shall be left after 10 minutes of inactivity or if a user logs in as Maintainer or Crew.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.658	The TOD shall be in Crew in crew compartment mode when the user has touched the TOD in the crew compartment. The mode shall be left after 10 minutes of inactivity or if the user logs in as Maintainer or Driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.659	The TOD shall be in Maintainer in crew compartment mode when the user has logged in as Maintainer on the TOD in the crew compartment. The mode shall be left after 10 minutes of inactivity or if the user logs in as Driver or Crew.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.371	When the TOD is in Idle mode no access to menus shall be given.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.372	When the TOD is in Driver in activated cab mode the TOD shall allow access to menus according to the menu tree of the HMI of the VZI-14.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

2F_10.03.Zefiro-Europe.CONCEPT.660	When the TOD is in Crew in activated cab mode the TOD shall allow access to menus according to the menu tree of the HMI of the VZI-14.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.661	When the TOD is in Maintainer in activated cab mode the TOD shall allow access to menus according to the menu tree of the HMI of the VZI-14.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.373	When the TOD is in Driver in non activated cab mode the TOD shall allow access to menus according to the menu tree of the HMI of the VZI-14.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.662	When the TOD is in Crew in non activated cab mode the TOD shall allow access to menus according to the menu tree of the HMI of the VZI-14.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.663	When the TOD is in Maintainer in non activated cab mode the TOD shall allow access to menus according to the menu of the HMI of the VZI-14.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.375	When the TOD is in Driver in crew compartment mode the TOD shall allow access to menus according to the menu tree of the HMI of the VZI-14.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.664	When the TOD is in Crew in crew compartment mode the TOD shall allow access to menus according to the menu tree of the HMI of the VZI-14.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.665	When the TOD is in Maintainer in crew compartment mode the TOD shall allow access to menus according to the menu tree of the HMI of the VZI-14.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

4.3 Menus

4.3.1 Primary TOD menus

4.3.1.1 *Login menu*

4.3.1.1.1 Function Design

To access functionality of the TOD a login ID is required. With login ID different user groups and languages are distinguished.
The user groups are:
* Driver
* Crew
* Maintainer
The supported languages are:
* Italian
* French
* German

* English
When a user logs in the language that is defined for that user is displayed.

The language displayed can be changed through the Settings menu.
To access functionality of the TOD a login ID is required. With login ID different user groups and languages are distinguished.
The user groups are:
* Driver
* Crew
* Maintainer
The supported languages are:
* Italian
* French
* Spanish
* English
When a user logs in, the language that is defined for that user is displayed.

The language displayed can be changed through the Settings menu.

4.3.1.1.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.21
Description	Access to the TOD shall be protected via a login procedure.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.56	The Login menu shall provide means to enter a numeric login id.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.400	The TOD in activated cab shall display the Train Inauguration menu when the cab is activated.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.401	The TOD in crew compartment shall display the Activation menu when the TOD is in Idle mode and the screen is touched. The TOD in non activated cab shall display the Login menu when the TOD is in Idle mode and the screen is touched.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.812	A login code and specific access level for Maintainers shall be provided on the TOD. Maintenance related menus shall only be shown and active in Maintainer mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_10.03.Zefiro-Europe.TRS.217
Description	The TOD shall support multiple language (namely French, German, Italian, English) selectable during the login.
Safety level	N/A

ID	2F_10.03.Zefiro-Europe.TRS.277
Description	The TOD shall support multiple language (namely French, Spanish, Italian, English) selectable during the login.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.399	The TOD shall be able to display Italian, French, German and English language.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.1007	The TOD shall be able to display Italian, French, Spanish and English language.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.402	The language shall be selected automatically when the correct login id has been entered.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.403	The displayed language shall be the language that is defined for the login id that has been entered.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

ID	2F_10.03.Zefiro-Europe.TRS.218
Description	Italian shall be the default language.
Safety level	N/A

ID	2F_10.03.Zefiro-Europe.TRS.278
Description	Spanish shall be the default language.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.405	Italian shall be the default language. Texts that are displayed before the login id has been confirmed shall be in Italian.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.1009	Spanish shall be the default language. Texts that are displayed before the login id has been confirmed shall be in Spanish.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

4.3.1.2 *Activation menu*

4.3.1.2.1 Function Design

When the cab has been activated the Train Inauguration menu is displayed. It presents the train configuration in a graphical format.

The activation menu shall be displayed after confirm of Train Inauguration and can also be reached through the main menu. it presents the status of the main components related to activation of the train.
The menu tree how to navigate to the Activation menu from the main menu has to be the same of the HMI of VZI-14.

4.3.1.2.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.59
Description	The activaion menu shall present the activation status of the train in a graphical format.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.62	The Activation menu shall display the complete train in a graphical format.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.184
Description	The activation menu shall present to the driver the status of the main components related to activation of the train.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.408	The Activation menu shall present to the Driver, status of the main components related to the activation of the train.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.1.3 *Main menu*

4.3.1.3.1 Function Design

The main menu provides the user access to all other menus. The Main menu consist of a set of buttons functionally grouped. The groups are diagnostics, instrumentation, settings and test.

4.3.1.3.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.63
Description	The main menu shall act as the menu used as the main access point for navigating the TOD structure.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.68	The main menu shall have navigation buttons, to access other menus.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.474	The user shall reach all menus from the main menu or subsequent menus to the main menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.475	The navigation buttons on the main menu shall be divided into 4 groups: Diagnostics Instruments Settings Test	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.1.4 *Station menu*

4.3.1.4.1 Function Design

The Station menu will be displayed automatically in the activated cab when train is at station. It contains information and functionality needed when the train is at station.
The Station menu will be displayed automatically in the activated cab when train is at station and the exterior doors are released. It contains information and functionality needed when the train is at station.

4.3.1.4.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.67
Description	The station menu shall present the door status to the driver when stopping at a station
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.470	The station menu shall automatically be shown on the TOD when the train is at station and the exterior doors are released.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.418	The station menu shall present status of exterior doors as follows. Closed Faulty Released Isolated (Door system not active)	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.991	The station menu shall present status allow manual control and provide acces to functions needed when coming to station.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1033	The station menu shall present status, allow manual control and provide access to functions needed when the train is at station and the exterior doors are released.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.121
Description	Corridor configuration: The station menu shall allow cut-out of doors.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.471	Corridor configuration: It shall only be possible to perform isolation (cut-out) of door if door is closed.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 2
2F_10.03.Zefiro-Europe.CONCEPT.473	Corridor configuration: It shall be possible to order isolate (cut-out) and activate (cut-in) of all passenger doors in a car. By selecting one car all doors, on both sides, in that car shall be affected by the cut-in or cut-out command.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 2

4.3.1.5 *Driving menu*

4.3.1.5.1 Function Design

The Driving menu is automatically displayed when the train speed is over standstill. The Driving menu shows a minimum of information in order not to distract the Driver of the train.

4.3.1.5.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.71
Description	The driving menu shall only present and allow access to necessary information and functions when driving the train.

Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.80	The Driving menu shall automatically be shown when speed of the train changes from stand still to speed.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.990	The driving menu shall only present information and allow access to necessary information and function when driving the train	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2 Menus for Diagnostics

4.3.2.1 *Events*

Events generated from various systems on the train are stored in the train diagnostics system (TDS). The TOD retrieves the event information from the TDS and displays it to the user. Events in the TDS are grouped and categorized.

Events are divided into the following categories and priorities.

1 - Serious fault events	Events that are stored in parallell to occurence of a serious fault.*) The ODBS event will give the driver added information to the serios fault..
2 - Advisory	Immediate action has to be taken by the driver. Indicates handling error.
3 - Alarm	Immediate action has to be taken by the driver or crew. Indicates train malfunction or substantial impact on train operation.
4 - Warning	Action has to be taken by the driver or crew. Indicates train malfunction or limited impact on train operation.
5 - Maintenance	Action has to be taken by maintenance staff. Indicates train malfunction.
6 - Other	No action has to be taken.

*) When a serious fault occurs a lamp on the driver's desk is lit and the fault is acknowledge using a button on the TOD frame. In parallel to that an event is also stored in the ODBS. The serious fault event is announced on the TOD and gives added information to the driver and it is acknowledged on the TOD display itself in the same way as Alarm and Warnings. Serious fault function is described in the CTC SFRS FA020023107chapter 3.1.13.
Events of categories 1, 3 and 4 occur it is presented to the user for acknowledge. The event is regardless of acknowledge, available for diagnostic purpose in the active event list menu, if it is still active, and in the event history menu.
Advisory events are not acknowledged, instead the user corrects the faulty handling and the Advisory event is then cleared by the corrective action.
Maintenance and Other events are not acknowledged and are only displayed to Maintainer in the active event menu or event history menu.

Prio	Category	Acknowledge	Active event overview menu	Active event menu	Event history menu
1	Serious fault event	YES	D/C/M	D/C/M	D/C/M
2	Advisory	*	*	*	D/C/M

3	Alarm	YES	D/C/M	D/C/M	D/C/M
4	Warning	YES	D/C/M	D/C/M	D/C/M
5	Maintenance	NO	M	M	M
6	Other	NO	NO	NO	M

* Advisory event is cleared when user has corrected the handling error

D = Driver, C = Crew, M = Maintainer

Serious fault events have the highest priority and takes precedence over all other category of events even if an event of another category has already been announced.

Advisory events have the second highest priority, and takes precedence over Alarm and Warning events even if an Alarm or Warning already has been announced.

Alarm events have the third highest priority and takes precedence over Warning events even if a Warning already has been announced.

4.3.2.2 *Events*

Events generated from various systems on the train are stored in the train diagnostics system (TDS). The TOD retrieves the event information from the TDS and displays it to the user. Events in the TDS are grouped and categorized.

Events are divided into the following categories and priorities.

1 - Serious fault events	Events that are stored in parallell to occurence of a serious fault.*) The ODBS event will give the driver added information to the serios fault..
2 - Advisory	Immediate action has to be taken by the driver. Indicates handling error.
3 - Alarm	Immediate action has to be taken by the driver or crew. Indicates train malfunction or substantial impact on train operation.
4 - Warning	Action has to be taken by the driver or crew. Indicates train malfunction or limited impact on train operation.
5 - Maintenance	Action has to be taken by maintenance staff. Indicates train malfunction.
6 - Other	No action has to be taken.

*) When a serious fault occurs a lamp on the driver's desk is lit and the fault is acknowledge using a button on the TOD frame. In parallel to that an event is also stored in the ODBS. The serious fault event is announced on the TOD and gives added information to the driver and it is acknowledged on the TOD display itself in the same way as Alarm and Warnings. Serious fault function is described in the CTC SFRS chapter 3.1.13.
Events of category 1, 3 and 4 occur it is presented to the user for acknowledge. The event is regardless of acknowledge, available for diagnostic purpose in the active event list menu, if it is still active, and in the event history menu.
Advisory events are not acknowledged, instead the user corrects the faulty handling and the Advisory event is then cleared by the corrective action.
Maintenance and Other events are not acknowledged and are only displayed to Maintainer in the active event menu or event history menu.

Prio	Category	Acknowledge	Active event overview menu	Active event menu	Event history menu
1	Serious fault event	YES	D/C/M	D/C/M	D/C/M
2	Advisory	*	*	*	D/C/M
3	Alarm	YES	D/C/M	D/C/M	D/C/M
4	Warning	YES	D/C/M	D/C/M	D/C/M
5	Maintenance	NO	M	M	M
6	Other	NO	NO	NO	M

* Advisory event is cleared when user has corrected the handling error

D = Driver, C = Crew, M = Maintainer

Serious fault events have the highest priority and takes precedence over all other category of events even if an event of another category has already been announced.

Advisory events have the second highest priority, and takes precedence over Alarm and Warning events even if an Alarm or Warning already has been announced.

Alarm events have the third highest priority and takes precedence over Warning events even if a Warning already has been announced.

4.3.2.3 *Active event overview menu*

4.3.2.3.1 Function Design

The active event overview menu provides a clear overview of the systems on the train that have active events. It also provides navigation possibilities to select systems with active events. With navigation it is meant that the active event list is shown, and contains active events for selected system.

4.3.2.3.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.47
Description	The active event overview menu shall provide an clear overview of which systems on the train have active events and also provide navigation possibilities for the systems with active alarms/faults.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.492	The active event overview menu shall present if there are any active events for the systems on the train. Systems available are: High voltage Battery supply Auxiliary supply Control and communication Traction Fire detection	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

	Air condition INFO Bogie Air supply Front Other Brakes Sanitary Doors N.B. The other group indicates active events for other systems on the train that are not mentioned in the other systems				
2F_10.03.Zefiro-Europe.CONCEPT.1026	The active event overview menu shall present if there are any active events for the systems on the train. Systems available are: High voltage Battery supply Auxiliary supply Control and communication Traction Fire detection Air condition INFO Bogie Air supply Front Other Brakes Sanitary Doors ATP system N.B. The other group indicates active events for other systems on the train that are not mentioned in the other systems	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.494	Selecting a system with active events shall bring up the active event list menu on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.902	Only systems with active events shall be possible to select in the active event overview menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.903	The active event list shall only contain active events for selected system when active event list is invoked from the active event overview menu..	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.496	Driver and Crew shall have active events of category Alarm and Warning indicated.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1012	Driver and Crew shall have active events of category Serious fault event, Alarm and Warning indicated.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

2F_10.03.Zefiro-Europe.CONCEPT.609	Maintainer shall have active events of categories Alarm, Warning and Maintenance indicated.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1013	Maintainer shall have active events of categories Serious fault event, Alarm, Warning and Maintenance indicated.	Derived Requirement	4S_09.03.05.-.TCM S Software	TOD	SIL 0

4.3.2.4 *Active event menu*

4.3.2.4.1 Function Design

The active event menu displays a scrollable list of currently active events. There are also a possibility to upon selection of a specific event, navigate to the expanded event menu.

4.3.2.4.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.75
Description	The active event menu shall present the active events on a train for the driver, crew or maintainer.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.610	The active event menu shall if there has been a system selection performed in the active event overview menu, present only active events for the selected system and selected area.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.503	There shall be a navigation button in the event field of the TOD that enables the user to go directly to the active event overview menu when threre is an active event.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.506	Driver and crew shall have active events of category Alarm and Warning displayed in the active event menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1014	Driver and crew shall have active events of category Serious fault event, Alarm and Warning displayed in the active event menu.	Derived Requirement	4S_09.03.05.-.TCM S Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.507	Maintainer shall have active events of category Alarm, Warning and Maintenance displayed in the active event menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1015	Maintainer shall have active events of category Serious fault event, Alarm, Warning and Maintenance displayed in the active event menu.	Derived Requirement	4S_09.03.05.-.TCM S Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.508	The active event menu shall be updated automatically if new events occur or existing events goes inactive.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.193

Description	The active event menu shall in a clearly arranged format (e.g. list format) present the active events for the selected area.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.427	For each active event in the list, key event properties shall be presented: Car Date/Time Code Event description	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.92	The active event menu shall in a list present the active events for the selected area.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.504	It shall be possible to scroll up and down in the active events list	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.505	It shall be possible to select an event in the list and receive expanded information about the event.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.5 *Event history overview menu*

4.3.2.5.1 Function Design

The event history overview menu is used to make a selection to only see history events for a specific event group.

4.3.2.5.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.205
Description	The event history overview menu shall provide selection and navigation possibility to view event history for a selected system.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.104	The event history overview menu shall provide selection buttons to the different systems that are available, as well a possibility to show all systems. High voltage Battery supply Auxiliary supply Control and communication Traction Fire detection Air condition INFO Bogie Air supply Front Other Brakes Sanitary Doors	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

2F_10.03.Zefiro-Europe.CONCEPT.1027	The event history overview menu shall provide selection buttons to the different systems that are available, as well a possibility to show all systems. High voltage Battery supply Auxiliary supply Control and communication Traction Fire detection Air condition INFO Bogie Air supply Front Other Brakes Sanitary Doors ATP system	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.6 *Event history menu*

4.3.2.6.1 Function Design

The event history menu displays a scrollable list of historical events on the train. There are also a possibility to upon selection of a specific event, navigate to the expanded event menu. Events of the Advisory type are considered historical when they have become inactive, i.e. the erroneous handling have been corrected by the user. Alarm, Warning, Maintenance and Other events are historical in the same moment they are generated and stored in the TDS.
The event history menu displays a scrollable list of historical events on the train. There are also a possibility to upon selection of a specific event, navigate to the expanded event menu. Events of the Advisory type are considered historical when they have become inactive, i.e. the erroneous handling have been corrected by the user. Serious fault event, Alarm, Warning, Maintenance and Other events are historical in the same moment they are generated and stored in the TDS.

4.3.2.6.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.84
Description	The event history menu shall present the event history on a train for the driver, crew or maintainer.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.530	It shall be possible to select to view the event history for the selected area.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.687	The event history menu shall if there has been a system selection performed in the history overview menu, present only events for the selected system and selected area.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.534	It shall be possible to initiate a new gathering of the event history, i.e. refresh.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.280
Description	The event history menu shall present history events from category 1 - 5 for the maintainer.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.535	Maintainer shall have historical events of all categories displayed in the event history menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.279
Description	The event history menu shall present history events from category 1 - 4 for driver and crew.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.668	Driver shall have historical events of categories Advisory, Alarm and Warning displayed in the event history menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1017	Driver shall have historical events of categories Serious fault event, Advisory, Alarm and Warning displayed in the event history menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.676	Crew shall have historical events of categories Advisory, Alarm and Warning displayed in the event history menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1018	Crew shall have historical events of categories Serious fault event, Advisory, Alarm and Warning displayed in the event history menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.191
Description	The event history menu shall in a clearly arranged format (e.g. in a list format) present the history of the events for the selected area.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.433	For each event in the list key data shall be presented: Car Date/Time Code Event description	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.110	The event history menu shall in a list present historical events.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.531	It shall be possible to scroll up and down in the event history list	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.533	It shall be possible to select an event in the list and receive expanded information about the event.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

4.3.2.7 *Expanded event menu*

4.3.2.7.1 Function Design

4.3.2.7.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.255
Description	The expanded event menu shall present detailed information of the selected event.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.755	The expanded event menu shall give the user detailed information of a selected event. The event information presented is: Category Time of activation Time of deactivation System Car identifier Car number Event identifier Process identifier Event message Driver text Repair text	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A
2F_10.03.Zefiro-Europe.CONCEPT.735	The Repair text in the expanded event menu shall only be present for the Maintainer.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	N/A

4.3.2.8 *Events*

4.3.2.8.1 Function Design

4.3.2.8.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.93
Description	The event menus shall present vital event related information to the driver
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.967	When an Serious fault event occurs it shall be presented in the event field of the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.122	When an event of the type Advisory is active, the Advisory shall be presented in the event field of the TOD in the activated cab.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.545	When an event of the type Alarm occurs, the Alarm shall be presented in the event field of the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.546	When an event of the type Warning occurs, the Warning shall be	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

	presented in the event field of the TOD.				

ID	2F_10.03.Zefiro-Europe.TRS.197
Description	Events shall be displayed differently depending of the priority of the event. Higher priority events always take precedence over an event of lower priority.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.968	The Serious fault event type shall have the highest priority when displayed on the TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.436	The Advisory event type shall have the second highest priority when displayed on the TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.547	The Alarm event type shall have the third highest priority when displayed on the TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.548	The Warning event type shall have the lowest priority when displayed on the TOD	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.688	The oldest unacknowledged event shall be presented first unless an event with higher priority occurs.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.8.3 Serious fault events

4.3.2.8.3.1 Function Design

When an Serious fault event has occurred the event is announced on the TOD to give added information to the serious fault that has occurred.

The Serious fault event is only announced on the TOD in active cab and it needs to be acknowledged. The event is available in the event history menu, and in the active event list if the serious fault event is still active.

The Serious fault event is presented in the event field of the TOD. If the event field is pressed, the Expanded event menu is shown presenting the user more information about the Serious fault event.

Note. Serious fault is acknowledged using the button on the frame of the TOD, see chapter 3.3.2.8. The serious fault event that is stored in the ODBS in parallell is also acknowledge, after the user has read it, using the acknowledge button on the TOD display.

4.3.2.8.3.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.272
Description	The Serious fault event shall be presented in the active cab when an Serious fault event has occurred.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.976	If a Serious fault event occurs the Serious fault event shall be shown on the TOD in the active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.273

Description	The Serious fault event shall be removed when the driver has acknowledged the event using the Acknowdge pushbutton on the display.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.977	The Serious fault event shall remain shown until the event is acknowledged by the Driver or Maintainer in active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.978	It shall be possible to acknowledge the Serious fault event.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.8.4 Advisory events

4.3.2.8.4.1 Function Design

When an Advisory event has occurred, immediate action needs to be taken by the user to correct the handling error. When the handling error is corrected the Advisory event is cleared, and it is available in the event history menu. The Advisory event is only announced on the TOD in active cab.
In the non-active cab or crew compartment it is possible to see in the active event menu if an Advisory event is active. The Advisory event is presented in the event field of the TOD. If the Advisory in the event field is pressed, the Expanded event menu is shown presenting the user more information about the Advisory event.
When an Advisory event has occurred, immediate action needs to be taken by the user to correct the handling error. When the handling error is corrected the Advisory event is cleared, and it is available in the event history menu. The Advisory event is only shown on the TOD in active cab.
The Advisory event is presented in the event field of the TOD. If the Advisory in the event field is pressed, the Expanded event menu is shown presenting the user more information about the Advisory event.

4.3.2.8.4.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.208
Description	The Advisory event shall be presented in the active cab when the driver has done an incorrect manuevre.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.979	If an Advisory event occurs the Advisory event shall be shown on the TOD in the active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.213
Description	The Advisory event shall be removed when the driver has corrected the faulty manuevre.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.439	The Advisory event shall remain shown until the Advisory event is no longer active.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.8.5 Alarm events

4.3.2.8.5.1 Function Design

When an Alarm event has occurred immediate action has to be taken by the user, due to a train malfunction or impact on train operation that is substantial. The Alarm event is announced on the TOD in active cab and/or in the crew compartment.The Alarm events need to be acknowledged by the user.

The event is available in the event history menu, and in the active event list if the alarm is still active.

The Alarm event is presented in the event field of the TOD. If the Alarm in the event field is pressed, the Expanded event menu is shown presenting the user more information about the Alarm event.

4.3.2.8.5.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.210
Description	The Alarm event shall be presented in the active cab and/or in the crew compartment when an Alarm event has occurred.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.274	If an Alarm event occurs the Alarm event shall be shown on the TOD in the active cab and/or in the crew compartment.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.214
Description	The Alarm event shall be removed when the user has acknowledged the event using the Acknowdge pushbutton on the display.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.442	The Alarm event shall remain shown until the event is acknowledged by the user.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.562	It shall be possible to acknowledge the Alarm event.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.563	Maintenance personnel shall be able to acknowledge all Alarm events at once.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.8.6 Warning events

4.3.2.8.6.1 Function Design

When a Warning has occurred immediate action has to be taken by the user, due to a train malfunction or impact on train operation that is limited. The Warning event is announced on the TOD in active cab and/or in the crew compartment.The Warning events need to be acknowledged by the user.

The event is available in the event history menu, and in the active event list if the warning is still active.

The Warning event is presented in the event field of the TOD. If the Warning in the event field is pressed, the Expanded event menu is shown presenting the user more information about the Warning event.

4.3.2.8.6.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.212
Description	Warning event information shall be presented on the lower part of the screen area of the display in the active cab and/or in the crew compartment when a Warning event has occurred.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.280	If a Warning event occurs the Warning event shall be shown in the event field of the TOD in the activated cab and/or in the crew compartment.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.215
Description	The Warning event information shall be removed when the user has acknowledged the event using the Acknowdge pushbutton on the display.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.445	The Warning event shall remain shown until the Warning is acknowledged.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.564	It shall be possible to acknowledge the Warning event.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.565	Maintenance personnel shall be able to acknowledge all Warning events at once.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.9 *Serious fault acknowledge*

4.3.2.9.1 Function Design

When a serious fault has occurred, this is indicated by a lamp on driver's desk and is acknowledged using the button on the TOD frame.

The function of the serious fault indication is described in Conventional Train Control SFRS, FA020023107, chapter 3.1.13.
When a serious fault has occurred, this is indicated by a lamp on driver's desk and is acknowledged using the button on the TOD frame.

The function of the serious fault indication is described in Conventional Train Control SFRS, chapter 3.1.13.

4.3.2.9.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.267
Description	The serious fault indication on driver's desk shall be acknowledged using the acknowledge button on the TOD frame.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.923	The serious fault indication on driver's desk shall be acknowledged using the acknowledge button on the TOD frame.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.10 *System messages*

4.3.2.10.1 Function Design

When a system message is displayed on the TOD it covers the entire TOD screen, and the functionality of the TOD application is blocked. I.e. the underlying state causing the system message to be displayed need to be deactivated in order to remove the system message from the TOD.

4.3.2.10.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.97
Description	System message menus shall present information to the user e.g. when the user attempts to activate the cab while another cab is already active or when shutting down the TOD.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.128	There shall exist system message menus for the following type of system events. - Power on - Communication failure - Another cab already active - Shut down - Waiting for train configuration - Decoupling - Lamp test	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.575	The system message menu shall cover the whole TOD screen area.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.576	The system message menu shall disappear when the system event is deactivated	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.10.3 Train configuration

4.3.2.10.3.1 Function Design

The train configuration system message is displayed when a decoupling of a train set has occurred and at each cab activation. The system message prompts the user to acknowledge the train configuration.

4.3.2.10.3.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.260
Description	At train inauguration the train configuration menu shall be displayed. When the user has acknowledged the train configuration the menu disappears.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.836	When a new train configuration has been loaded the train configuration menu shall be displayed. On the menu the train configuration to confirm shall be presented.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 2
2F_10.03.Zefiro-Europe.CONCEPT.992	After the cab has been activated the train configuration menu shall be displayed on the TOD in the activated cab.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.896	If the safe acknowledge button is pressed the new configuration shown on TOD shall be accepted.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 2
2F_10.03.Zefiro-Europe.CONCEPT.897	Once the driver has acknowledged the new configuration the menu shall disappear.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

2F_10.03.Zefiro-Europe.CONCEPT.904	The train configuration system message menu shall not be possible to leave unless acknowledged..	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.10.4 Lamp Test

4.3.2.10.4.1 Function Design

The Lamp test system message is displayed on the TOD when the user has pressed the lamp test push button on the Driver's desk. The system message gives the user ability to detect pixel faults on the TOD.

4.3.2.10.4.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.262
Description	At lamp test the TOD shall display a white menu covering the entire screen so that the user can detect if there are any pixel faults.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.835	When lamp test action has been ordered from active driver desk the TOD shall show a white screen for pixel testing.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.906	The Lamp test system message menu shall not be displayed if Decoupling system message menu is active and displayed.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.907	The Lamp test system message menu shall not be displayed if Fire menu is active and displayed.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.10.5 Power On

4.3.2.10.5.1 Function Design

This system message is displayed when the TOD is powered up and TOD application is starting.

4.3.2.10.5.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.220
Description	When the TOD unit is powered on a startup menu shall be displayed indicating that the application is starting
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.289	When the TOD is starting up from it´s power off state a system message menu shall be presented on the TOD informing the user about this	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.10.6 Communication failure

4.3.2.10.6.1 Function Design

The system message that is shown on the TOD when the communication with the train computer is lost. I.e. the lifesign from the train computer is no longer valid.

4.3.2.10.6.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.222
Description	A communication error message menu shall be displayed if the communication to the CCU in the leading cab is lost.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.295	If communication between the TOD and the train computer is lost a system message menu shall be presented on the TOD informing the user about this.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.960	There shall be a possibility to store the start time and end time of the communication lost state duration in the TCMS event database.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	TBD

4.3.2.10.7 Another cab already active

4.3.2.10.7.1 Function Design

When a user tries to activate another cab while there still is an active cab in the train, this system message is displayed on the TOD.

4.3.2.10.7.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.224
Description	Another cab already active message menu shall be displayed if an attempt to activate the cab is made when another cab is already active.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.301	If a cab is active and another cab is activated the TOD shall present a system message menu in the cab that is trying to become active, informing the user that only one cab can be active in the train	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.10.8 Shut down

4.3.2.10.8.1 Function Design

When a shutdown order from the train computer is issued the Shutdown system message is show on the TOD.

4.3.2.10.8.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.226
Description	A shut down message menu shall be displayed when the train is ordered to shut down.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_10.03.Zefiro-Europe.CONCEPT.307	When the train is ordered to shut down to it´s power off state a system message menu shall be presented on the TOD informing the user about this	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.2.10.9 Waiting for train configuration

4.3.2.10.9.1 Function design

When reconfiguration of the train topology has occurred, and configuration is not valid, the Waiting for train configuration system message is shown on TOD.

4.3.2.10.9.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.253
Description	A waiting for train configuration message menu shall be displayed when the train configuration can not be retrieved or is not valid.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.738	When train configuration not can be retrieved, a system message menu shall be presented on the TOD informing the user about this.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3 Menus for Instruments

4.3.3.1 *Traction-brake menu*

4.3.3.1.1 Function Design

This menu shows the traction and brake levels per car.

4.3.3.1.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.51
Description	The traction-brake menu shall present the current traction/brake status in a graphical format and allow cut-out and cut-in of dynamic brake
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.134	The Traction-brake menu shall display the status of traction and brake per car as bars in a chart.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.949	There shall be a possibility to navigate to the Traction-Brake menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.958	There shall be a possibility to perform cut in and cut out of dynamic brake.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.2 *Interlocks menues*

4.3.3.2.1 Function Design

The Interlocks menus is a set of menus with an overview menu on top. The overview menu will indicate if there exists any interlocks. Upon active interlock of some kind, it is possible to navigate further down in the Interlocks menu structure to get more detailed information about the interlocks currently active.

4.3.3.2.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.101
Description	The interlocks menu shall present on TOD if there are any interlocks resulting in traction block, emergency brake or speed limitation
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.140	The Interlocks menu shall consist of one overview menu and sub-menus for traction block, speed limit and emergency brake.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.552	The interlock overview page shall have navigation buttons for the sub-menus.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.553	The navigation buttons shall have the colour grey when no interlock is active and red when at least one interlock is active for that sub-menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.554	The traction block interlock menu shall present all possible causes for traction block to be active.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.556	The emergency brake interlock menu shall present all possible causes for emergency brake to be active.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.558	The speed limit interlock menu shall present all possible causes for speed limit to be active.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.560	The current speed limit shall be presented on the speed limit interlock menu in km/h	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.555	Each possible cause shall have a status indication : · Not active · Active	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.3 *System overview menu*

4.3.3.3.1 Function Design

The System overview menu is the top menu for easy navigation to the other instrumentation menus. E.g. Battery, Sanitary and Bogie menu.

4.3.3.3.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.109

Description	The system overview menu shall provide navigation buttons to all the system menus of the TOD.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.152	The following navigation buttons shall be available on the system overview menu: · High voltage · Traction · Bogie · Brakes · Battery · Fire detector · Air supply · Tank supervision · Coupler cover · Auxiliary HVAC Interior lighting	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.652	Those menus not listed under the system menu shall be accesible directly from the main menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.4 *High voltage system menu*

4.3.3.4.1 Function Design

The High voltage system menu provides functionality related to train's high voltage systems. E.g. cut out of pantographs and circuit breakers and disconnector status.
The High voltage system menu provides functionality related to train's high voltage systems. E.g. circuit breakers and disconnector status.

4.3.3.4.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.113
Description	The high voltage system menu shall present status and allow control of the high voltage system via TOD .
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.158	There shall be a possibility to navigate to the High voltage menu	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.569	Line voltage shall be presented on the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.5 *Traction system menu*

4.3.3.5.1 Function Design

The Traction system menu provides functionality related to train's traction systems. E.g. cut out orders and status of converter modules.

4.3.3.5.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.117
Description	The traction system menu shall present status and allow manual control of the traction system via TOD.

Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.164	There shall be a possibility to navigate to the Traction menu	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.981	The traction system menu shall present status and allow manual control of the traction system.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.189
Description	It shall be possible to cut-out and cut-in main components on the traction system menu (e.g. line converters).
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.980	It shall be possible to cut-out and cut-in main components on the traction system menu (e.g. line converters)	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.6 *HVAC system menu*

4.3.3.6.1 Function Design

The HVAC system menu is reached from the Main menu and displays information related to the HVAC system both in saloons and in cab. The menu also allows manoeuvring of the HVAC systems.
The HVAC system menu is reached from the System overview menu and displays information related to the HVAC system both in saloons and in cab. The menu also allows manoeuvring of the HVAC systems.

4.3.3.6.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.125
Description	The HVAC system menu shall present status and allow manual control of the HVAC system.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.680	It shall be a possible to navigate to the HVAC system menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.176	The HVAC system menu shall present status and allow manual control of the HVAC system.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.7 *Battery system menu*

4.3.3.7.1 Function Design

The Battery system menu provides functionality related to train's battery supply systems. E.g. status of battery chargers.

4.3.3.7.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.129
Description	The battery system menu shall present the battery system status on the TOD.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.951	There shall be a possibility to navigate to the Battery menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.8 *Brake system menu*

4.3.3.8.1 Function Design

The Brake system menu provides functionality related to train's brake systems. E.g. status of brake computer unit, parking brake and emergency brake.

4.3.3.8.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.133
Description	The brake system menu shall present status and allow to control of the brake system.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.952	There shall be a possibility to navigate to the Brake menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.989	The brake system menu shall present status and allow control of the brake system.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.9 *Tank supervision menu*

4.3.3.9.1 Function Design

The Tank supervision menu provides functionality related to tanks on the train and to the Toilet system. E.g. fresh water tank level, waste water tank level and ability to order drain of tanks.

4.3.3.9.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.137
Description	The Tank supervision menu shall present status and allow control of the Tanks on the train (toilet, water supply, waste water system).
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.953	There shall be a possibility to navigate to the Tank supervision menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.591	The tank supervision menu shall present status and allow control of the train (toilet, water supply, waste water system).	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.10 *Fire detection system menu*

4.3.3.10.1 Function Design

The Fire detection system menu provides functionality related to train's fire detection systems. E.g. status of the fire detectors and linear heat cable detectors, and location of detectors.

4.3.3.10.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.142
Description	The fire detection system menu shall present status and allow control of the fire detection system.
Safety level	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.281
Description	The fire detection system menu shall present the status of the fire detection system on the TOD.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.954	There shall be a possibility to navigate to the Fire detection menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.482	The Fire detector menu shall be automatically displayed if fire is detected.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.905	The Fire detector menu shall not be displayed at fire if Decoupling system message menu is active.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.11 *Bogie monitoring system menu*

4.3.3.11.1 Function Design

The Bogie system menu provides functionality related to train's bogie systems. E.g. bogie status and instability monitoring system status.

4.3.3.11.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.146
Description	The bogie monitoring system menu shall present status and allow control of the bogie monitoring system.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.955	There shall be a possibility to navigate to the Boogie menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.938	The bogie system menu shall present status and allow control of the bogie monitoring system.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.12 *Air supply system menu*

4.3.3.12.1 Function Design

The Air supply system menu provides functionality related to train's air supply systems. E.g. status of compressors and sensors, and performing cut out/cut in of compressors.

4.3.3.12.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.150
Description	The air supply system menu shall present status and allow control of the air supply system.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.956	There shall be a possibility to navigate to the Air supply menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.899	The TOD shall present status and allow control of the air supply system.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.13 *Interior lights menu*

4.3.3.13.1 Function Design

The Interior lights system menu is reached from the Main menu and displays information related to the Interior lights system. The menu also allows manoeuvring of the Interior lights systems.

4.3.3.13.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.276
Description	The interior lights system menu shall present status and allow manual control of the interior lights system.
Safety level	

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.988	There shall be a possibility to navigate to the Interior lights system menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.987	The Interior lights menu shall present status and allow manual control of the interior lights system	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.3.14 *Coupler cover menu*

4.3.3.14.1 Function Design

The Coupler cover menu provides functionality related to train's coupler cover device. E.g. status of the cover and manouvering.

4.3.3.14.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.154
Description	The front system menu shall present status of the air front couplers and front hatches.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.957	There shall be a possibility to navigate to the Coupler cover menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.775	Coupler cover manouvering shall only be possible to do from active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.4 Menus for Settings

4.3.4.1 *Settings menu*

4.3.4.1.1 Function Design

The Settings menu provides the user with settings related functionality. From this menu the user can proceed to e.g. the language selection menu and perform logout action.

4.3.4.1.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.55
Description	The Settings menu shall provide possibility to navigate to the different setting functions and also adjust the intensity of the TOD screen.
Safety level	SIL 0

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_10.03.Zefiro-Europe.CONCEPT.225	The settings menu shall provide navigation button to: · Master reset · Language menu	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.514	It shall be possible to view the current display intensity value.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.515	It shall be possible to adjust the display intensity.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.516	It shall be possible to log out from the TOD. After logout the login menu shall be presented.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.236
Description	It shall be possible to switch between languages in run-time.
Safety level	SIL 0

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_10.03.Zefiro-Europe.CONCEPT.454	There shall be a language selection menu from where the user can select a new language to be used for text presentation for the entire TOD application.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.909	It shall not be needed to restart the application in order to activate the new language.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.910	The possible languages to select shall be English, German, Italian and French.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.4.2 *Master reset menu*

4.3.4.2.1 Function Design

The Master reset menu allows the user to issue a train master reset order to the train computer to carry out.

4.3.4.2.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.158

Description	The Master reset menu shall allow activation of master reset on the train via TOD.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.231	It shall be possible to do master reset in Active cab	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.517	The master reset button shall be greyed out and de-activated when master reset is not allowed.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.813	A button on the TOD, only accessible in Maintainer mode, shall be provided to trigger a "master reset maintainer".	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.4.3 *Language menu*

4.3.4.3.1 Function Design

The Language menu allows the user to change the language to be used on the TOD menus. The language switch is performed at once and no restart of the TOD application needs to be performed.

4.3.4.3.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.164
Description	The country menu shall present status and allow to set country.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.237	Possible languages shall be: □□□□□□□□· English □□□□□□□□· Italian □□□□□□□□· French □□□□□□□□· German	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

ID	2F_10.03.Zefiro-Europe.TRS.236
Description	It shall be possible to switch between languages in run-time.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.682	The current selected language shall be indicated.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1036	There shall be a language selection menu from where the user can select a new language to be used for text presentation for the entire TOD application.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1037	It shall not be needed to restart the application in order to activate the new language.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1023	The possible languages to select shall be English, Spanish, Italian and French.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.4.4 *Program version menu*

4.3.4.4.1 Function Design

The Program version menu lists the software versions loaded on the various system components on the train. The menu is for Maintainers only. Sorting and filtering of the list can be applied.

4.3.4.4.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.166
Description	The program version menu shall present the program versions of TCMS and sub-system devices.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.243	The TOD shall in a list format present the system device's program versions.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.519	It shall be possible to scroll up and down in the program version list	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.888	It shall be possible to perform sorting and/or filtering of the program version list.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.520	For each program version in the list the following data shall be presented. · Unit · Type · Version	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.4.5 *PreHeating Menu*

4.3.4.5.1 Function Design

The Preconditioning Menu provides the user with ability to start/stop Preheating Mode.

4.3.4.5.2 Requirements

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.1000	It shall be possible to navigate to the preheating menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1001	The TOD shall display instructions for how to activate/deactivate the preheating mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1002	The TOD shall show the status of preheating mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.1003	The TOD shall show all info messages about Preconditioning mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.5 Menus for Test and Maintenance

4.3.5.1 *Test overview menu*

4.3.5.1.1 Function Design

The test overview menu provides the user with ability to start different train tests.

4.3.5.1.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.178
Description	The test overview menu shall allow activation of different applicable train tests.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.642	There shall be a possibility to navigate to the test overview menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.641	The test overview menu shall allow activation of train tests.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.5.2 *Master controller test menu*

4.3.5.2.1 Function Design

The Master controller test menu is used when performing testing of the master control handle.

4.3.5.2.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.174
Description	The master controller test menu shall allow activation of a master controller test, give advice how to perform the test and present the test result
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_10.03.Zefiro-Europe.CONCEPT.632	It shall be possible to navigate to the master controller test menu.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.633	The TOD shall show the result of the master controller test.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.634	It shall be possible to activate the master controller test from the TOD.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0
2F_10.03.Zefiro-Europe.CONCEPT.635	The TOD shall display instructions for how to perform the master controller test.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 0

4.3.5.3 *Brake test menu*

4.3.5.3.1 Function Design

The Brake test menu is used when performing brake tests. Instructions are presented on the menu how to perform the activated test.

4.3.5.3.2 Requirements

ID	2F_10.03.Zefiro-Europe.TRS.170
Description	The brake test menu shall allow activation of a brake test, give advice how to perform the test and present the test result via TOD.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_10.03.Zefiro-Europe.CONCEPT.624	The brake test menu shall allow activation of brake test, give information and display result of the brake test.	Derived Requirement	4S_09.03.05.-.TCMS Software	TOD	SIL 2

5 Signal Interface

5.1 I/O Interface

5.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_10.03.Zefiro-Europe.CONCEPT.28			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_10.03.Zefiro-Europe.CONCEPT.29			

5.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_10.03.Zefiro-Europe.CONCEPT.30			

Outputs:

5.2 Bus Interface

Supplier to TCMS SW:
TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_10.03.Zefiro-Europe.CONCEPT.33			